

# Using AI for Reproducible Research

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## A full research paper in 24 hours

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## What does research say (for the time being)?

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AI handles the technical work better, but struggles with ideas and research judgment.

### **AI-written papers are weaker in ideas than in execution**

AI models rated 912 AI-written economics papers and 41 published papers (Li, 2026)

They also rated scored equally high on execution, but fewer ideas reached top-journal level

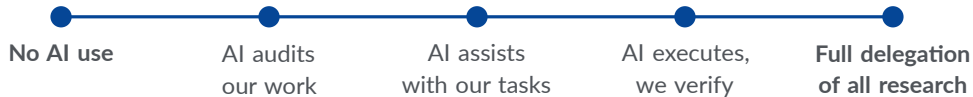
### **Authors reject the work produced by AI agents working on the same RQs**

AI agents had six days to answer the question of an unpublished paper (Kirgis et al., 2026)

The paper's authors reviewed the result: the conclusions did not follow from the evidence

## Where on this scale do I want to be?

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Reflect yourself while I explain my current position

## One fundamental challenge

### Research must be reproducible



Every run must return the same results

We must be able to verify every step

### LLMs are probabilistic



The same input can return different outputs

LLMs can report the same steps differently

**Problem!**

## To solve this, we need to understand a few basics about AI tooling

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### **Harness**

The software around an LLM that lets it read files, run code, and search the web (e.g., Claude Code, Codex, dsh, and PI)

### **Agent**

An LLM in a harness that works on a task independently and chooses its own steps

### **Skill**

A folder with written instructions and scripts that tell an agent how to do a specific task

### **AGENTS.md** (CLAUDE.md)

A text file with our rules that the agent reads at the start of every session

### **Other tooling**

MCP, evals, hooks, memory, and sandboxes

## What should this AI tooling do?

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1

### Clear objectives and processes

It should help us set well-defined goals and structured processes, which is what reproducible research requires: A clear methodological approach.

2

### Clear roles for humans and agents

It should optimize where human judgment is needed/valuable and where boring steps can be delegated to agents (i.e., no need for verification).

## Solution: skill-first over agent-first AI tooling

### Skill-first

Design workflows that we or our agents can call



*A workflow sets the steps and gates*

### Agent-first

Design autonomous agents with responsibilities



*The agent chooses the steps*

Skill-first tooling can ensure: researcher-in-the-loop.

## Simple example: the paper audit skill

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**Purpose:** Execute an audit (mistakes, errors, inconsistencies, and fact-check) in a paper before it goes to a journal.

- **Input:** I give it the paper, any other materials, and optional instructions
- **Plan:** The main model lists the tables, figures, and references, and the most likely errors
- **Execute:** Four helper models each check one part (references, tables, text, response letter)
- **Review:** The main model reviews the findings, looks for missed errors, and writes one report
- **Audit:** A helper model checks each finding against the paper, and the main model revises
- **Result:** I check the final report, which ranks each mistake with its location

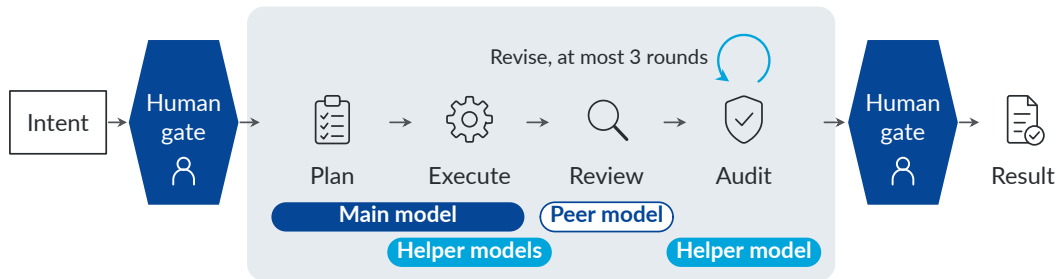
## More complex example: the paper reviser skill

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**Purpose:** Execute a preliminary revision of a paper based on comments.

- **Input:** I give it the paper, reviewer/editor/discussant comments, and optional instructions
- **Intent:** The main model writes how it reads my request, and I correct it (1st gate)
- **Plan:** The main model turns each comment into a task and sets the order of the changes
- **Execute:** The main model edits the paper and drafts preliminary responses to the comments, and a helper model verifies new references
- **Review:** A peer model reviews the revisions with fresh eyes, and the main model corrects
- **Audit:** A helper model checks that each comment is addressed correctly
- **Result:** I process each change, the response, and points left open (2nd gate)

## My current skill template (extended version)



See the [Anthropic documentation](#) and [agentskills.io](#).

## “Starter” skills repository

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Simple starter skills for writing a research paper and conducting research

### **Paper finder**

Careful search for research papers on a topic

### **English editor**

Corrects grammar and style, and marks every change

### **Feedback reviser**

Processes changes based on comments

### **Document auditor**

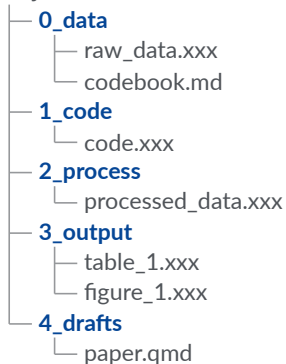
Checks every reference, fact, and number in a document

[github.com/victorvanpelt/research-skills](https://github.com/victorvanpelt/research-skills)



## But how can AI help us analyze data?

### Project folder



### Four rules

1. Never modify **0\_data**
2. Only save data to **2\_process**
3. Only load data from **0\_data** or **2\_process**
4. Anything in **2\_process** and **3\_output** can be deleted and regenerated from **1\_code**

### Agents have two roles:

- Role 1: Agents edit the code, or help us edit it
- Role 2: AI as a source of generated data only

## AGENTS.md: rules for every agent

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```
# AGENTS.md (excerpt, shortened)
- Never modify the raw data in 0_data/.
- Never edit 2_process/ or 3_output/ by hand.
- Change only 1_code/, 4_drafts/, and the codebook.
- Before a multi-file change, say which files will change.
- Never run git commit or git push.
- Done only when "make clean" then "make" rebuilds everything.
- Report what changed and anything you could not verify.
```

Git is a quick way to see which changes AI agents made (researcher-in-the-loop)

Source: [github.com/victorvanpelt/research\\_pipeline\\_example](https://github.com/victorvanpelt/research_pipeline_example)

## Recent development: System One models (e.g., Jev)

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A new development inspires a classification of models into two types, based on Kahneman's (2011) dual-system theory.

### System Two model



Good at complex tasks that need reasoning

### System One model



Good at fast and structured decisions

## System One models could be useful for research

In a chatbot experiment, we tell the chatbot how to behave in each condition, but we cannot be sure that every reply follows this instruction.

Thus, we let Jev classify how the chatbot behaves in each reply, either to create a quality score after the experiment or even to intervene before the participant sees the reply.

### Test on 802 synthetic chatbot replies: how often did Jev answer correctly?

| Instruction that Jev checked              | Replies that followed it | Replies that did not follow it | All replies |
|-------------------------------------------|--------------------------|--------------------------------|-------------|
| Rule for all participants                 | 99.1%                    | 100%                           | 99.3%       |
| No markdown or formatting                 | 644 of 650               | 152 of 152                     | 796 of 802  |
| Assertive treatment                       | 92.5%                    | 100%                           | 96.3%       |
| Direct and confident, without hedge words | 371 of 401               | 401 of 401                     | 772 of 802  |
| Non-assertive treatment                   | 98.0%                    | 99.8%                          | 98.9%       |
| Tentative and cautious, with hedge words  | 393 of 401               | 400 of 401                     | 793 of 802  |

## Final comments on AI use in research

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1

### Data protection

We should prefer a local LLM, then a cloud LLM with zero data retention (ZDR) in the EU, then other cloud LLMs. Some data cannot be shared with a cloud LLM at all.

2

### Self-disclosure of AI use

Most journals already require disclosure. It should be standard practice that we reflect on and are transparent about how we use AI, because this prevents misconceptions and promotes responsible use.

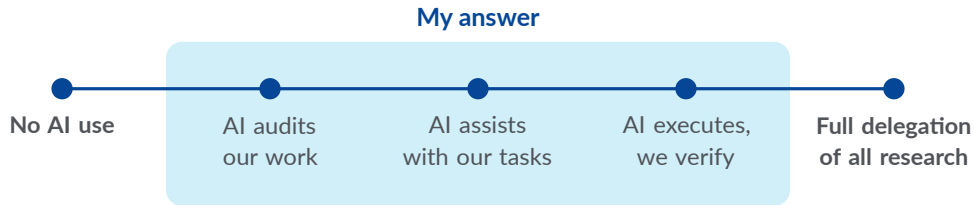
3

### Statistical inference

AI lowers the hurdles to run many non-hypothesized tests, fish for results, and exploit researcher degree's of freedom. Thus, preregistration and sharing research materials become more important than ever.

## My position on the scale

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Where on the scale are you / do you want to be?

## Thank you for your attention

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Slides: <https://github.com/victorvanpelt/ai-research-talk>

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## References

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